

Paul Abrams

Toronto, Ontario

Full-Stack Developer · Ruby on Rails · React & TypeScript · AI-Forward Engineering · Cloud, CI/CD & DevOps

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Professional Summary

Senior full-stack developer with 15+ years shipping web applications end-to-end — idea, design, build, deploy, iterate. **AI-forward by default:** daily, hands-on use of **Claude Code, GitHub Copilot, Cursor, and Gemini CLI** across multiple languages and stacks, with a current contract designing prompt fixtures and rubrics that **evaluate and refine AI-generated code** for correctness, security, and design quality — including a multi-week **Ruby on Rails** feature delivery against the **Discourse** codebase (production Ruby, RSpec test coverage, OmniAuth 2FA enforcement flow). Strong on the **React + TypeScript** front end (custom Azure DevOps web extensions, a chess training app on the Lichess REST API, an end-to-end multiplayer web game I shipped solo) and on the **server side** (n-tier ASP.NET serving 5,000+ users, Node.js/Express + WebSocket REST/event servers, PostgreSQL/SQL Server with concurrency-correct transactions and tuned execution plans). **Cloud & DevOps** fluency: CI/CD pipelines in **Azure DevOps** and **GitHub Actions**, **Docker** containerization, **Kubernetes** deployment, IAM/OAuth 2.0, TLS / secret hygiene, ITSG-33 / PBMM controls — Azure-anchored cloud experience with patterns that transfer cleanly to **AWS**. **Startup DNA:** since August 2025 I have been operating as an independent contractor, shipping features end-to-end on tight loops. **Professional Scrum Master I.** Active open-source contributor (60+ public PRs).

Professional Experience

Abrams Software Kingston / Toronto, Ontario

August 2025 – Present

Full-Stack Developer · Independent

- **AI as a daily productivity multiplier.** Hands-on, contract-level work with **Claude Code, GitHub Copilot, Cursor, and Gemini CLI** across **Ruby, TypeScript, React, C#, Python, Go, Rust, SQL**, and **VBA**. Routine workflow: scope a feature, prompt the model, review and refine the generated code, run it against tests, audit for security and performance, ship. Same loop the role describes — including **building and shipping features end-to-end** with AI as a core productivity multiplier.
- **Ruby on Rails feature delivery against Discourse.** Multi-week engagement writing production **Ruby** code and **RSpec** test coverage for a Discourse feature in the OmniAuth callbacks path — a setting (`enforce_second_factor_on_external_auth`) that lets administrators require 2FA for local logins while still permitting OAuth-provider sign-ins. Exercised the controller logic, locale files, and the full RSpec test suite for OAuth flows; ran the work inside Discourse’s **Docker**-heavy build and runtime environment. Engagement was for **Stellar.ai** as part of LLM evaluation work — I wrote the human reference implementation that competing model outputs were graded against.
- **LLM training and code evaluation** for [DataAnnotation.tech](https://dataannotation.tech) and Stellar.ai. Designed prompt fixtures and grading rubrics that probe **specific model failure modes** — **Ruby on Rails** feature deliveries with RSpec assertions; **security audits** at known-vulnerable revisions with CWE/CVSS-graded inventories; refactor tasks that grade for code *deletion* (not just additions); adversarial unit tests that cleanly separate models that fix implementations from ones that fix tests; cross-stack migrations; and AI-assistant “skills” that codify methodology and are evaluated by running the same task with and without them.
- **TWNR — architected and shipped a multiplayer trading game as a full-stack web application, solo.** **React**-style component patterns on the front end (TypeScript); **Express + WebSocket** REST/event server on the back; **PostgreSQL** persistence with concurrency-correct transactions (`SELECT ... FOR UPDATE`); **JWT auth**; JSON-Schema-generated REST API contract that doubles as live documentation. Idea → design → build → deploy in tight iterations. Live at twnr.paulabrams.ca.

- **chess.js x-ray attacker support** — diagnosed a missing feature with React-lifecycle breakpoint debugging, opened [upstream PR #572](#), and published a fork as [@pabrams/chess.js](#) on npm. Powers [Chess Attack Visualization](#), a **React + TypeScript** training app consuming the Lichess.org REST API.
- **60+ public PRs** to outside repos — Microsoft's `vsts-team-calendar` and `tfs-cli`, the official TypeScript-Website, the **Microsoft Authentication Library .NET** docs, `canada-ca/platform-security`, the Digma split-keyboard configurator, and others. Comfortable jumping into unfamiliar codebases — exactly the discipline a Rails monolith and a React front end both demand.

Employment and Social Development, Canada Gatineau, QC

December 2021 – July 2025

Senior Software Engineer, Infrastructure and Cloud Enablement (BDM Project)

- Enhanced and maintained a **microservices** enterprise onboarding application — **C# / .NET, Blazor, Azure SQL, Azure Functions, GraphQL**. Designed and consumed **REST APIs** between services; integrated with enterprise IAM via **Azure AD / Entra (OAuth 2.0, OpenID Connect)**; published and governed APIs through **Azure API Management** with authentication, throttling, and transformation policies — cloud patterns that map directly to **AWS API Gateway + Cognito / IAM**.
- Designed and shipped **custom Azure DevOps web extensions** in **React, TypeScript, and Node.js** — a build/deploy analytics widget that renders success-rate charts from the ADO Builds REST API; a people-by-project reporter that uses an in-memory hashmap to cross-reference user/project membership efficiently (the naive query pattern was unusably slow — a **performance** and **query-design** call); and a weekly work-item digest emailer driven by scheduled pipelines. Responsive UI, custom styling, accessible interactions.
- Built **PowerShell tooling** against the Azure DevOps REST API to snapshot area paths, iteration paths, security configurations, and org-level state on a schedule and commit those snapshots into Git as a **configuration-drift detector** — automation that closed a gap no off-the-shelf product covered. Same “ship the tool, don’t file the ticket” instinct the role asks for.
- **CI/CD pipelines** in Azure DevOps for build, test, **Docker** containerization, **Kubernetes** deployment, and Azure Data Factory (ETL) orchestration. Implemented **ITSG-33 (NIST-based) security controls** — TLS / SSL certificate management, secret hygiene via Azure Key Vault — for SA&A submissions against a PBMM security profile. Daily **Git** workflow.
- Ran weekly **demos and design walkthroughs** for junior developers on Azure DevOps, Key Vault, app registration, and custom extensions. Active participant in **daily stand-ups** with BAs, QA, and developers.

Employment and Social Development, Canada Gatineau, QC

August 2019 – November 2021

Development Lead, Testing Services

- Co-architected an **internal C# / .NET web application** enabling QA staff to author structured test scenarios via a web UI; designed the **SQL Server** schema and developed the code bridging the application with Azure DevOps via its **REST API**. Integrated with the departmental headless CMS platform.
- Engineered Azure DevOps **CI/CD pipelines** for **automated testing** — provisioned Windows agent machines to execute parallel **Selenium** UI tests against deployed applications and integrated logged test results back into ADO work items.
- Championed the departmental transition from **LoadRunner to JMeter**, reducing licensing costs while maintaining robust performance-testing capabilities. **Code review** and standards enforcement for the team.

Department of Immigration, Refugees, and Citizenship, Canada

Ottawa, ON

August 2017 – August 2019

Team Lead / Technical Lead

- Led **Agile / Scrum** sprint planning, review, retrospective, **daily stand-up**, and backlog refinement; pioneered capacity-planning and forecasting using Azure DevOps Server; introduced poker planning estimation.

- **Mentored developers in SQL Server query optimization** — query tuning, indexing strategy, partitioned tables, execution-plan analysis — against large TFS databases and warehouses. Also coached on SSRS and C# / .NET.
- Implemented and deployed **custom web extensions in React / TypeScript / Node.js** with responsive layouts.

Department of Justice, Canada Ottawa, ON

December 2015 – August 2017

Development Lead

- Dev lead on the **Legal Case Management System (iCase)** — a large n-tier **ASP.NET / C#** enterprise web application used by 5,000+ lawyers — while concurrently building its **Microsoft Dynamics CRM** successor. Modified iCase schema and front-end UI to surface migration status to users in real time.
- Engineered MS Dynamics CRM **plug-ins, custom actions, and workflow processes in C#**; rewrote SQL stored procedures as C# with LINQ; used SSIS for ETL between iCase and Dynamics. Applied design patterns (Repository, Strategy, Mediator) and clean-code refactoring. **Deployments to Windows servers** via both manual runbooks and scripted pipelines.
- Established **code-review processes** integrated directly with TFS work items — gating standards for maintainability, security, and performance — and mentored junior developers.

Department of Citizenship and Immigration, Canada Ottawa, ON April 2014 – December 2015

Senior Software Engineer

- Hired alongside one other engineer to take over a **TFS / TFVC** release-management environment from a contractor team. **Decommissioned over-engineered legacy C# desktop release-management apps** and replaced them with lightweight PowerShell scripts calling the **TFS REST APIs** — net result, most of the contractor tooling was simply deleted. **Bias for action and ownership.**
- Implemented and deployed Team Foundation Server **web extensions in React / TypeScript / Node.js**; maintained a fork of Microsoft's vsts-team-calendar extension with configurable colours, Outlook integration, release-management features, and IE-compatibility shims. UI design and styling work end-to-end.
- Guided development teams in migrating from legacy XAML builds to modern, scriptable build definitions; **deployed across Windows environments** both manually and via automated pipelines.

Department of Justice, Canada Ottawa, ON

October 2008 – April 2014

Full-Stack Developer (2009–2014) · Software Development Engineer in Test (2008–2009)

Core contributor to the department's flagship **Legal Case Management System (iCase)** — a distributed multi-tier **ASP.NET / C#** enterprise web application used by 5,000+ lawyers and agents across Canada. Multi-tier with role- and object-based security, integrated with records management, financial, and email systems, 24/7 high availability.

Full-Stack Developer (2009–2014)

- Led development of the **"My iCase" widget portal** — the new entry point to iCase. Ran JAD sessions, produced prototypes, authored BDD/TDD documentation, and implemented the application largely solo. **ASP.NET Web Parts and the Personalization framework** with a **custom SQL personalization provider, Knockout.js MVVM** on the front end (a direct ancestor of the **React** patterns I use today), async AJAX web-service calls, async HTTP handlers, ReportViewer in remote SSRS mode.
- Diagnosed a homepage report taking ~10 s uncached: profiled with **SQL Profiler, Report Execution logs, and actual execution plans**, then replaced live joins with a flat table populated by a nightly job. **~50x faster uncached, 3-4x faster cached.** Textbook **query-tuning through execution-plan analysis.**
- Built a **self-initiated availability-monitoring suite** for web and document servers because support staff were running checks manually evenings and weekends; eliminated significant overtime and reduced operational risk.
- Diagnosed and resolved a **COM Interop memory leak** on the document servers; built automated health monitoring to catch regressions.

Software Development Engineer in Test (2008–2009)

- Identified a critical regression-testing bottleneck and voluntarily architected a custom framework in **IBM Rational Functional Tester** that parsed English-language scenario documents and auto-generated executable test scripts.
- Replaced the department's **“all-hands” manual regression cycle** (~20–25 testers running for two weeks every release) with one person reviewing automated results in five days. **~98% reduction in release validation effort**, eliminated the standing 2-week code freeze per release, **saved an estimated \$160,000 per cycle**.

Education

Bachelor of Computer Science (Honours) - Carleton University

September 1999 – May 2003

Certifications & Awards

Professional Scrum Master I (Scrum.org) · ITIL v3 Foundation (Loyalist Certification Services) · DOJ Team Merit Award 2012 (Cost Recovery Process Improvement); Team Merit Award 2011 (Chart of Accounts); Team Spirit Award 2009